

t-distributed Stochastic Neighbor Embedding (t-SNE)

(1–15)

1. Q: What does the acronym t-SNE stand for?

A: **t-distributed Stochastic Neighbor Embedding.**

2. Q: Is t-SNE a linear or non-linear dimensionality reduction method?

A: **Non-linear.**

3. Q: What is the primary goal of t-SNE?

A: To map high-dimensional data to a low-dimensional space while **preserving local similarities** between points.

4. Q: In t-SNE, which distribution is used to quantify similarity between points in high-dimensional space?

A: **Gaussian distribution** (or normal distribution).

5. Q: Write the formula for the conditional probability ($P_{j|i}$).

A:

$$P_{j|i} = \frac{\exp(-|\vec{x}_i - \vec{x}_j|^2 / 2\sigma_i^2)}{\sum_{k \neq i} \exp(-|\vec{x}_i - \vec{x}_k|^2 / 2\sigma_i^2)}$$

6. Q: The parameter (σ_i) is related to which hyperparameter?

A: **Perplexity**, which controls the effective neighborhood size.

7. Q: What is the symmetrized probability (P_{ij}) formula?

A:

$$P_{ij} = \frac{P_{j|i} + P_{i|j}}{2n}$$

8. Q: In low-dimensional space, which distribution is used to quantify similarity?

A: **Student's t-distribution with 1 degree of freedom.**

9. Q: Why use Student's t-distribution in low-dimensional space?

A: To **reduce the crowding problem** and allow moderate distances to be more distinguishable.

10. Q: Which two distributions are compared using KL divergence in t-SNE?

A: The **high-dimensional probability distribution (P)** and the **low-dimensional probability distribution (Q)**.

11. Q: Write the formula for KL-divergence minimized in t-SNE.

A:

$$KL(P||Q) = \sum_{i \neq j} P_{ij} \log \frac{P_{ij}}{Q_{ij}}$$

12. Q: Which optimization algorithm is used in t-SNE?

A: Gradient descent (usually with momentum).

13. Q: What data characteristic is successfully preserved by t-SNE?

A: Local neighborhood structure (local similarities).

14. Q: Main limitation of t-SNE regarding data structure?

A: Poor preservation of global structure; distances between clusters may not be meaningful.

15. Q: Symbolic expression for similarity (q_{ij}) in low-dimensional space?

A:

$$q_{ij} = \frac{(1 + \|\vec{y}_i - \vec{y}_j\|^2)^{-1}}{\sum_{k \neq l} (1 + \|\vec{y}_k - \vec{y}_l\|^2)^{-1}}$$

II. Uniform Manifold Approximation and Projection (UMAP) (16–30)

16. Q: What does UMAP stand for?

A: Uniform Manifold Approximation and Projection.

17. Q: What did t-SNE generally fail to preserve that UMAP aims to preserve?

A: Global data structure (relationships between clusters).

18. Q: First step in UMAP regarding each datapoint?

A: Determine **nearest neighbors** for each point.

19. Q: Which distance metric is used to find neighbors in UMAP?

A: Typically **Euclidean distance**, but UMAP can support other metrics.

20. Q: How is the distance to the nearest neighbor (ρ_i) defined?

A: As the **minimum distance** from point (i) to any other point to avoid zero distances.

21. Q: Formula for the probability (p_{ji}) of point (j) being a neighbor of (i)?

A:

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$$p_{j|i} = \frac{\exp\left(-\frac{\max(0, d(x_i, x_j) - \rho_i)}{\sigma_i}\right)}{\sum_j \exp\left(-\frac{\max(0, d(x_i, x_j) - \rho_i)}{\sigma_i}\right)}$$

22. Q: Main statistical technique UMAP uses to preserve structure?

A: **Fuzzy topological representation** using a **fuzzy simplicial set**.

23. Q: UMAP minimizes cross-entropy between what?

A: The **high-dimensional fuzzy simplicial set** and the **low-dimensional embedding**.

24. Q: Two matrices/functions for cross-entropy in UMAP?

A: **High-dimensional adjacency (similarity) matrix** and **low-dimensional similarity matrix**.

25. Q: Primary advantage of UMAP over t-SNE regarding embedding consistency?

A: UMAP produces **more stable and reproducible embeddings**.

26. Q: Computational cost comparison of UMAP vs t-SNE?

A: **UMAP is faster**, especially for large datasets.

27. Q: In which mathematical field is UMAP more theoretically grounded than t-SNE?

A: **Topological data analysis / manifold theory**.

28. Q: How does Student's t-distribution in t-SNE help address crowding?

A: Its **heavy tails** spread out moderately distant points in low-dimensional space.

29. Q: Typical low-dimensional space (Q dimensions) for visualization?

A: **2D or 3D** for plotting.

30. Q: Which method is preferred when global structure preservation is critical?

A: **UMAP**.